THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 6	Common prime factors give HCF $= 6$
2	**(a)** 6	Substitute $x = 4$: $p(4) = 6$
3	**(a)** infinitely many solutions (coincident lines)	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$
4	**(c)** $x^2 + 3x + 2 = 0$	Degree must be 2
5	**(b)** 4	$a_2 - a_1 = 4$
6	**(d)** 5	$\sqrt{3^2 + 4^2} = 5$
7	**(b)** 5	$\sqrt{x^2 + y^2}$
8	**(c)** $\frac{4}{5}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
9	**(b)** 7 m	$h = 7\tan 45^\circ = 7 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 7$ m
10	**(c)** 4 cm	$PT^2 = OP^2 - OT^2 = 5^2 - 3^2 = 16$; $PT = 4$ cm
11	**(c)** 154 cm²	$\frac{22}{7} \times 7 \times 7$; Area $= 154$ cm^2
12	**(c)** 220 cm²	$2 \times \frac{22}{7} \times 7 \times 5$; CSA $= 220$ cm^2
13	**(a)** 15	$\frac{10+20}{2}$; Class mark $= 15$
14	**(b)** $\frac{1}{2}$	Favourable outcomes: 2, 4, 6; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
15	**(d)** 12	LCM $= \dfrac{4 \times 6}{2} = 12$
16	**(d)** 4	$2x - 8 = 0 \Rightarrow x = 4$
17	**(a)** $(3, 2)$	$2(3) + 3(2) = 12 = 12$
18	**(d)** 2	At $x = 2$ the left side equals 0, so 2 is a root
19	**(a)** 39	$a + (n-1)d = 39$
20	**(a)** 3	Perimeter ratio = side ratio

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (1); HCF $= 6$ (1).	1+1
22	$p(3) = 10$ (2).	2
23	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$ (1); all ratios equal, infinitely many solutions, consistent (1).	1+1
24	Put $x = 2$: LHS $= 0$ (1); equals RHS 0, so it is a root (1).	1+1
25	$5, 8, 11, 14, \ldots$ (2).	2
26	$h^2 = 36 + 64 = 100$ (1); $h = 10$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$26 = 2 \times 13$, $91 = 7 \times 13$ (1); HCF $= 13$ (1); LCM $= 182$ (1).	1+1+1
28	Factors $(x-3)(x-5)$ (1); zeroes $3, 5$ (1); sum $8 = -b/a$ verified (1).	1+1+1
29	Multiply (i) by -1 and (ii) by 1 and subtract: $(-2)x = -10$ (2); $x = 5$ (1); put in (i): $y = 3$ (1); check in (ii) (1).	2+1+1+1
30	Split the middle term: $-5 = -3 - 2$ because $ac = 6$ (1); regroup and factorise (1); roots $x = 2, 3$ (1).	1+1+1
31	$a = 3$, $d = 4$ (1); $a_n = a + (n-1)d$ (1); $a_{12} = 3 + 11(4) = 47$ (1).	1+1+1
	OR: $h^2 = 3^2 + 4^2 = 25$ (1); $h = 5$ cm (1); perimeter $= 12$ cm (1).	1+1+1
32	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(3)^2 + (4)^2} = \sqrt{25}$ (1); $= 5$ units (1).	1+1+1
	OR: $AC = \sqrt{3^2 + 4^2} = 5$ cm (1); $\sin A = \frac{BC}{AC} = \frac{4}{5}$, $\cos A = \frac{AB}{AC} = \frac{3}{5}$ (1); $\tan A = \frac{BC}{AB} = \frac{4}{3}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$135 = 3^3 \times 5$, $225 = 3^2 \times 5^2$ (2); HCF $= 45$ (1); LCM $= 675$ (1); $45 \times 675 = 30375$ verified (1).	2+1+1+1
34	Zeroes 2, 3 (2); sum $= 5 = -\frac{b}{a}$ (1.5); product $= 6 = \frac{c}{a}$ (1.5).	2+1.5+1.5
35	Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x = -21$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
36	Split the middle term: $-7 = -6 - 1$ because $ac = 6$ (1); regroup and factorise (1); roots $x = \frac{1}{2}, 3$ (1).	1+1+1
	OR: $a = 7$, $d = 3$, $l = 64$ (1); $l = a + (n-1)d$ gives $n - 1 = 19$ (1); $n = 20$ (1).	1+1+1
37	Ladder is the hypotenuse of a right triangle (1); $d^2 = 5^2 - 4^2 = 9$ (1); $d = 3$ m (1).	1+1+1
	OR: Area $= \frac{1}{2}\lvert x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\rvert$ (1); $= \frac{1}{2}\lvert 0 + 12 + 0\rvert$ (1); $= 6$ sq. units (1).	1+1+1
38	Hypotenuse $= \sqrt{4^2 + 3^2} = 5$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1).	1+1+1
	OR: Distances of the points from the foot: $h\cot 30^\circ = 18$ m and $h\cot 60^\circ = 6$ m (2); The points are on opposite sides, so the distances add (2); Distance $= 24$ m (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** $(4, 6)$	Average the coordinates
2	**(d)** $\frac{4}{3}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
3	**(d)** $5\sqrt{3}$ m	$h = 5\tan 60^\circ = 5 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 5\sqrt{3}$ m
4	**(c)** 6 cm	$r^2 = OP^2 - PT^2 = 10^2 - 8^2 = 36$; $r = 6$ cm
5	**(b)** 44 cm	$2 \times \frac{22}{7} \times 7$; Circumference $= 44$ cm
6	**(d)** 1540 cm³	$\frac{22}{7} \times 7^2 \times 10$; Volume $= 1540$ cm^3
7	**(c)** 30–40	It belongs to the class 30–40; Modal class $=$ 30–40
8	**(a)** $\frac{1}{3}$	Favourable outcomes: 3, 6; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
9	**(c)** 3	$12 = 2^2 \times 3$
10	**(d)** 5	Sum of zeroes $= -b/a$
11	**(b)** 7	$k = 2(2) + 3(1)$
12	**(c)** -3	Compare with $ax^2 + bx + c = 0$
13	**(d)** 55	$S_n = \frac{n}{2}[2a + (n-1)d]$
14	**(c)** 6	$\frac{AD}{DB} = \frac{AE}{EC}$
15	**(c)** first quadrant	Signs of coordinates
16	**(d)** $\frac{1}{2}$	Standard value
17	**(c)** $3\sqrt{3}$ m	$h = 9\tan 30^\circ = 9 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 3\sqrt{3}$ m
18	**(a)** 25 cm	$OP^2 = r^2 + PT^2 = 7^2 + 24^2 = 625$; $OP = 25$ cm
19	**(a)** 7 cm	$r = \frac{44 \times 7}{2 \times 22}$; $r = 7$ cm
20	**(a)** 25 cm	$l^2 = 7^2 + 24^2 = 625$; $l = 25$ cm

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(3)^2 + (4)^2} = \sqrt{25}$ (1); $= 5$ units (1).	1+1+1
22	$\sin\theta = \frac{4}{5}$ (1) $= \frac{4}{5}$ (1).	1+1
23	Let the height be h; in the right triangle, $\tan 45^\circ = \frac{h}{10}$; $h = 10\tan 45^\circ = 10 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$) (1); Height $= 10$ m (1).	1+1
24	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 25^2 - 20^2 = 225$ (1); $PT = 15$ cm (1).	1+1
25	Area of a circle $= \pi r^2$; $\frac{22}{7} \times 3.5 \times 3.5$ (1); Area $= 38.5$ m^2 (1).	1+1
26	Total surface area $= 2\pi r(r + h)$; $2 \times \frac{22}{7} \times 7 \times (7 + 10)$ (1); TSA $= 748$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Ladder is the hypotenuse; height $= 12\sin 60^\circ$, base $= 12\cos 60^\circ$ (1); $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ (1); Height $= 6\sqrt{3}$ m, distance $= 6$ m (1).	1+1+1
28	$PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(16) + 2(12)$ (1); Perimeter $= 56$ cm (1).	1+1+1
29	Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{45}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 16.5$ cm (1).	1+1+1
30	Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 7^2 \times 24$ (1); Volume $= 1232$ cm^3 (1).	1+1+1
31	Class marks x_i: 5, 15, 25, 35, 45 (1); $\sum f_i = 50$ and $\sum f_i x_i = 970$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{970}{50} = 19.4$ (1).	1+1+1
	OR: Total tickets $= 10 + 25 = 35$ (1); Favourable tickets (prizes) $= 10$ (1); $P = \frac{10}{35} = \frac{2}{7}$ (1).	1+1+1
32	HCF $= 6$, LCM $= 60$ (1+1); HCF×LCM $= 360 = 12 \times 30$ (1).	1+1+1

	OR: $p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (5)x + (6)$ (1); one such polynomial is $x^2 - 5x + 6$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{5^2 - 3^2} = 4$ cm; perimeter $= 2(4 + 3)$ (2); Perimeter 14 cm; area $= 2 \times \frac{1}{2} \times 3 \times 4 = 12$ cm^2 (1).	2+2+1
34	Outer radius $= 21$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(21^2 - 14^2)$ (2); Area of path $= 770$ m^2 (1).	2+2+1
35	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24\times12\times6}{3^3}$ (2); $n = 64$ (1).	2+2+1
36	Take assumed mean $a = 85$; deviations $d_i = x_i - a$: -30, -20, -10, 0, 10, 20 (2); $\sum f_i = 50$ and $\sum f_i d_i = -510$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 85 + \frac{-510}{50} = 74.8$ (1).	2+2+1
	OR: Total balls $= 10$ (2); Favourable balls: 3, 6, 0 (2); Probabilities: $\frac{3}{10}$, $\frac{3}{5}$, 0 (1).	2+2+1
37	$12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (3); HCF $= 6$ (1); LCM $= 72$ (1).	3+1+1
	OR: Sum $= 6 = -\frac{b}{a}$ (2); sum of products in pairs $= 11 = \frac{c}{a}$ (2); product $= 6 = -\frac{d}{a}$ (1).	2+2+1
38	Let pencil $= x$, eraser $= y$: $2x + 3y = 22$, $3x + 2y = 23$ (2); solving, $x = 5$, $y = 4$ (2); pencil ₹5, eraser ₹4 (1).	2+2+1
	OR: Breadth x, length $x + 5$: $x(x + 5) = 66$ (1); $x^2 + 5x - 66 = 0$ (1); $(x - 6)(x + 11) = 0$ (1); $x = 6$ m (breadth), length 11 m (reject negative) (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 40–50	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class = 40–50
2	**(b)** $\frac{1}{4}$	Favourable outcomes: HH; number = 1; $P = \frac{1}{4}$
3	**(c)** $\sqrt{7}$	7 is not a perfect square, so $\sqrt{7}$ is irrational
4	**(b)** 6	Product of zeroes $= c/a$
5	**(d)** 7	$x + y = 10$, $x - y = 4$ gives $2x = 14$
6	**(a)** 2 and 3	Put each factor equal to zero
7	**(c)** $4, 7, 10, 13, \ldots$	Only $4, 7, 10, 13, \ldots$ has the same difference 3 between consecutive terms
8	**(d)** 3	Side ratio is the square root of the area ratio
9	**(c)** $(3,\ 0)$	$y = 0$ on the x-axis
10	**(d)** 1	Identity
11	**(c)** 9 m	$x = 9\cot 45^\circ$; Shadow = 9 m
12	**(b)** 14 cm	Perimeter $= PA + AO + OB + BP = 2(4) + 2(3)$; Perimeter = 14 cm
13	**(b)** 7 cm	$r^2 = \frac{154 \times 7}{22} = 49$; $r = 7$ cm
14	**(b)** 550 cm²	CSA $= \pi rl$; CSA $= \frac{22}{7} \times 7 \times 25 = 550$ cm^2
15	**(c)** 50	$3 + 9 + 4 + 18 + 16$; $\sum f_i = 50$
16	**(d)** $\frac{3}{8}$	Favourable balls = 3; $P = \frac{3}{8}$
17	**(d)** $2^2 \times 3 \times 5$	Divide repeatedly by primes: $60 = 2^2 \times 3 \times 5$
18	**(a)** 2	A polynomial of degree n has at most n zeroes
19	**(c)** y-axis	All points have the same x-coordinate
20	**(a)** 1	$D = b^2 - 4ac$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Class marks x_i: 15, 25, 35, 45; $\sum f_i = 50$ and $\sum f_i x_i = 1740$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1740}{50} = 34.8$ (1).	1+1
22	Total possible outcomes = 6: 1, 2, 3, 4, 5, 6; Favourable outcomes: ; number = 0 (1); $P = \frac{0}{6} = 0$ (1).	1+1
23	Divide repeatedly by primes (1); $140 = 2^2 \times 5 \times 7$ (1).	1+1
24	$3x - 12 = 0$ (1) so $x = 4$ (1).	1+1
25	Adding: $2x = 16$ (1) so $x = 8$, $y = 4$ (1).	1+1
26	$x - 4 = 0$ or $x - 5 = 0$ (1); $x = 4$ or $x = 5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	From (i): $y = \dfrac{18 - 2x}{3}$ (1); substitute in (ii) (1); simplify to $(1)x = 3$ (1); $x = 3$ (1); $y = 4$ (1).	1+1+1+1+1
28	$a = 2$, $b = -4$, $c = 3$ (1); $D = -8$ (1); no real roots (1).	1+1+1
29	$a = 2$, $d = 3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{10} = \frac{10}{2}[4 + 9(3)] = 155$ (1).	1+1+1
30	By the Basic Proportionality Theorem, $\dfrac{AD}{DB} = \dfrac{AE}{EC}$ (1); $\dfrac{2}{3} = \dfrac{4}{EC}$ (1); $EC = 6$ cm (1).	1+1+1
31	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (5,\ 5)$ (1).	1+1
	OR: Base $= \sqrt{5^2 - 4^2} = 3$ (1); $\cos A = \frac{3}{5}$ (1); $\tan A = \frac{4}{3}$ (1).	1+1+1
32	Angle of depression = angle of elevation of the top of the cliff from the boat $= 60^\circ$ (1); $\tan 60^\circ = \frac{10\sqrt{3}}{x}$ so $x = 10\sqrt{3}\cot 60^\circ$ (1); Distance = 10 m (1).	1+1+1

	OR: $\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 3 \times 4$ (1); Area $= 12$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$a_n = 6 + 19(4) = 82$ (1); $S_n = \frac{20}{2}(6 + 82) = 880$ (2).	1+2
34	The triangles formed are similar (AA) (1); $\frac{H}{6} = \frac{30}{4}$ (1); $H = 45$ m (1).	1+1+1
35	Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (2, 3)$ (2).	1+2
36	$PR = 10$ cm (1); $\sin P = \frac{4}{5}$, $\cos P = \frac{3}{5}$, $\tan P = \frac{4}{3}$ (2); $\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2 = 1 = 1$ (2).	1+2+2
	OR: Let height $= h$; the distances from the foot are $h \cot 30^\circ$ and $h \cot 60^\circ$, which differ by 6 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 6$ gives $h = 6 \div \left(\frac{2\sqrt{3}}{3}\right)$ (2); Height $= 3\sqrt{3}$ m (distances 9 m and 3 m) (1).	2+2+1
37	The chord touches the smaller circle at M, so $OM \perp$ chord and M is the midpoint of the chord (2); Half-chord $= \sqrt{25^2 - 7^2} = 24$ cm (2); Chord $= 2 \times 24 = 48$ cm (1).	2+2+1
	OR: The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 7^2 = 154$ cm^2; square $= 196$ cm^2 (2); Left over $= 42$ cm^2 (1).	2+2+1
38	Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(6)^3}{\frac{4}{3}\pi(2)^3} = \left(\frac{6}{2}\right)^3$ (2); $n = 27$ (1).	2+2+1
	OR: Cumulative frequencies: 10, 12, 18, 34, 40; $N = 40$, so $\frac{N}{2} = 20$ (2); Median class 40–50: $l = 40$, $cf = 18$, $f = 16$, $h = 10$ (2); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{20-18}{16} \times 10 = 41.25$ (1).	2+2+1

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